1. Derivatives: A Deeper Look

This reading reinforces the core derivative concepts and focuses on building intuition for the applications you found challenging.

1.1. The Derivative as Rate of Change

The derivative $f'(x)$ measures the **instantaneous** rate of change of f at x. Geometrically, it's the slope of the tangent line. But more importantly, it answers:

"At this exact moment, how fast is the quantity changing?"

This interpretation is crucial for related rates problems.

1.2. Related Rates: The Strategy

Related rates problems involve quantities that change **together** over time. The key insight:

1. Identify what you're given ($\frac{dx}{dt}$, known values)
2. Identify what you want ($\frac{dy}{dt}$)
3. Find a **geometric relationship** connecting x and y
4. Differentiate that relationship with respect to time t
5. Substitute and solve

The hard part is often step 3 — finding the right equation. Common relationships:

- **Pythagorean:** $x^2 + y^2 = z^2$ (ladder, distance)
- **Volume:** $V = \pi r^2 h$ (conical tank, cylinder)
- **Similar triangles:** Maintains proportions as things scale
- **Area:** $A = \pi r^2$, $A = xy$

Key insight: Draw the diagram. Label what's changing and what's constant.

1.3. Optimization: Finding Extrema

Optimization means finding maximum or minimum values of a function.

1.3.1. The Procedure

1. Identify the quantity to optimize (call it Q)
2. Express Q as a function of **one variable** (use constraints to eliminate others)
3. Find critical points: $Q'(x) = 0$ or where $Q'(x)$ DNE
4. Classify critical points (first or second derivative test)
5. Check endpoints if on a closed interval
6. Verify your answer makes physical sense

1.3.2. Common Pitfalls

- Forgetting domain restrictions
- Not checking endpoints on closed intervals
- Finding critical points but not verifying they're max/min
- Algebraic errors when simplifying the objective function

1.4. Building Intuition

1.4.1. Why the Chain Rule Matters

When variables depend on each other, changes propagate. The chain rule captures this:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

In related rates, we're often given $\frac{dx}{dt}$ and need $\frac{dy}{dt}$. The chain rule bridges them.

1.4.2. Implicit Differentiation Recap

When y is defined implicitly by $F(x, y) = 0$:

1. Differentiate both sides with respect to x
2. Remember: $\frac{d}{dx}[y] = \frac{dy}{dx}$ (chain rule!)
3. Collect $\frac{dy}{dx}$ terms and solve

Example: $x^2 + y^2 = 25$

- Differentiate: $2x + 2y\frac{dy}{dx} = 0$
- Solve: $\frac{dy}{dx} = -\frac{x}{y}$

1.5. Reflection Questions

1. In related rates, why must we differentiate **before** substituting numerical values? What goes wrong if we substitute first?

2. A common optimization pattern: "Find the maximum area of a rectangle with fixed perimeter." Why does the solution always turn out to be a square? What does this tell you about symmetry in optimization?

3. The derivative $f'(x)$ gives the rate of change. What does $f''(x)$ tell you about how that rate itself is changing? How is this useful in curve sketching?

4. L'Hôpital's rule requires an indeterminate form. Why can't we apply it to $\lim_{x \to 0} \frac{\sin x}{x+1}$? What is the actual limit?

5. Draw a function that is: (a) increasing and concave up, (b) increasing and concave down, (c) decreasing and concave up. How do f' and f'' differ in each case?